

AgriVisionAI — Technical Project Documentation

Problem Statement ID: skh005

Problem Statement: Smart Agricultural Market Intelligence System

Team: The Elite Four

Team ID: 8K5YVT

1. Project Overview

AgriVisionAI is an AI-driven agricultural market intelligence and decision-support system that helps farmers decide when and where to sell. It combines crop-price forecasting with weather, seasonal demand and transportation-cost factors to estimate maximum expected profit.

Problem Statement

- Farmers may lack reliable information about the best market for a crop.
- Future crop prices are difficult to estimate using manual judgement.
- Different mandis can offer different prices, while transport cost changes actual net profit.
- Unanalysed decisions and guesswork can reduce realised farm income.

Proposed Solution

- Collect market, weather and location data.
- Forecast crop-price movement using AI/ML.
- Rank mandis using transport-adjusted expected profit.
- Generate Sell Now / Wait and best-mandi recommendations.
- Support multilingual delivery and device-agnostic SMS/IVR/missed-call delivery.

Technical Overview

Reference stack: Flutter, FastAPI/Python, PostgreSQL, XGBoost, Prophet, scikit-learn, Google Maps API, OpenWeather API, Agmarknet, data.gov.in, eNAM, Firebase Cloud Messaging, JWT and Google Translate.

2. System Architecture & Technology Stack

Architecture

Farmer/Basic Phone → Flutter or SMS/IVR layer → FastAPI backend → Decision Engine → PostgreSQL + external market/weather/maps data → Recommendation + notifications.

Key Modules

- Client interface; authentication; data ingestion; AI decision engine; profit optimiser; notification layer; database.

Technology Stack

Layer	Technology
Frontend	Flutter
Backend	Python + FastAPI
Database	PostgreSQL
AI/ML	XGBoost, Prophet, scikit-learn
Weather	OpenWeather API
Market	Agmarknet, data.gov.in, eNAM
Maps	Google Maps API
Notifications	Firebase Cloud Messaging
Auth	JWT
Language	Google Translate API

System Requirements

- Android/mobile or web-capable client; Python/FastAPI server; PostgreSQL; external API/data connectivity; SMS/IVR gateway for basic-phone delivery.

3. System Design

Data Flow

1) Farmer inputs crop, quantity and location → 2) market/weather data collected → 3) forecasting → 4) mandi comparison → 5) net-profit ranking → 6) Sell/Wait recommendation → 7) app/SMS/IVR delivery.

Core Data Entities

Entity	Representative fields
Farmer	farmer_id, phone, language, location
Crop	crop_id, crop_name, season
MarketPrice	crop_id, mandi_id, date, price
Weather	location, timestamp, conditions
Mandi	mandi_id, name, latitude, longitude
Transport	origin, destination, distance, cost
Prediction	crop_id, forecast_date, predicted_price
Recommendation	farmer_id, action, best_mandi, expected_profit
Notification	recipient, channel, language, timestamp

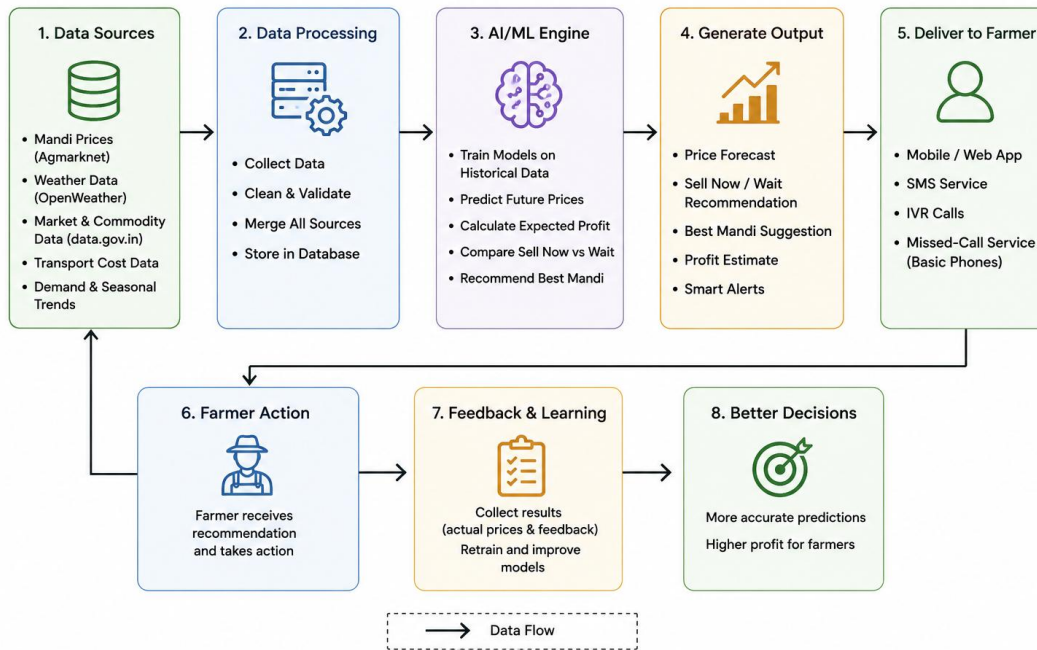
Roles & Interfaces

- Farmer: inputs and recommendations. FPO/agriculture department: institutional use. Admin: configuration and data operations.
- External interfaces: Agmarknet/data.gov.in/eNAM, OpenWeather, Google Maps, Firebase; SMS/IVR gateway for basic phones.

4. Technical Workflow & Methodology

End-to-End Workflow

AgriVisionAI – Technical Workflow



Core Logic

Expected Net Profit = Expected Sale Value – Estimated Transportation Cost – Other applicable selling costs.

Best Mandi = $\text{argmax}(\text{Net Profit})$ over eligible mandis.

Sell/Wait threshold should be calibrated using validation data; the supplied deck does not specify a fixed threshold.

Model / Methodology

- Prophet: time-series trend/seasonality component.
- XGBoost: nonlinear feature-based prediction component.
- scikit-learn: preprocessing, utilities and evaluation.

Final model selection should be based on out-of-sample validation.

5. Implementation

Module-wise Implementation

Module	Implementation
Flutter UI	Login, crop/quantity/location, recommendation and language controls.
FastAPI	REST services for auth, data retrieval, prediction and recommendations.
PostgreSQL	Users, market/weather records, predictions and recommendations.
Prediction service	Python pipeline wrapping selected forecasting/ML model.
Mandi ranking	Distance + transport cost + price → net-profit ranking.

Notifications	Firestore push plus external SMS/voice gateway.
Language	Google Translate API; voice layer can connect to speech/telephony services.

Important Details

- Normalise crop/mandi names, dates, units and price fields.
- Handle missing data through cleaning/interpolation.
- Use cached CSV/offline data for API failure.
- Use offline cache and synchronisation for network issues.
- Monitor prediction error and retrain/validate continuously.

Device-Agnostic Decision Output

Keep the recommendation channel-neutral: crop, action, best mandi, expected profit, validity and explanation. Render the same payload in the app or convert it to SMS, IVR or missed-call responses.

7. Results, Limitations & Future Scope

Results / Outputs

- AI-assisted Sell Now / Wait recommendation.
- Best mandi ranked using expected profit and transport cost.
- Multilingual and voice/text-oriented accessibility.
- Potential reduction in unnecessary transport and improved decision quality.

Technical Limitations

- Forecast quality depends on data quality and coverage.
- External API outages affect real-time inputs.
- Sparse data can reduce model reliability.
- Transport cost varies with actual local conditions.
- SMS/IVR/missed-call delivery needs gateway integration.

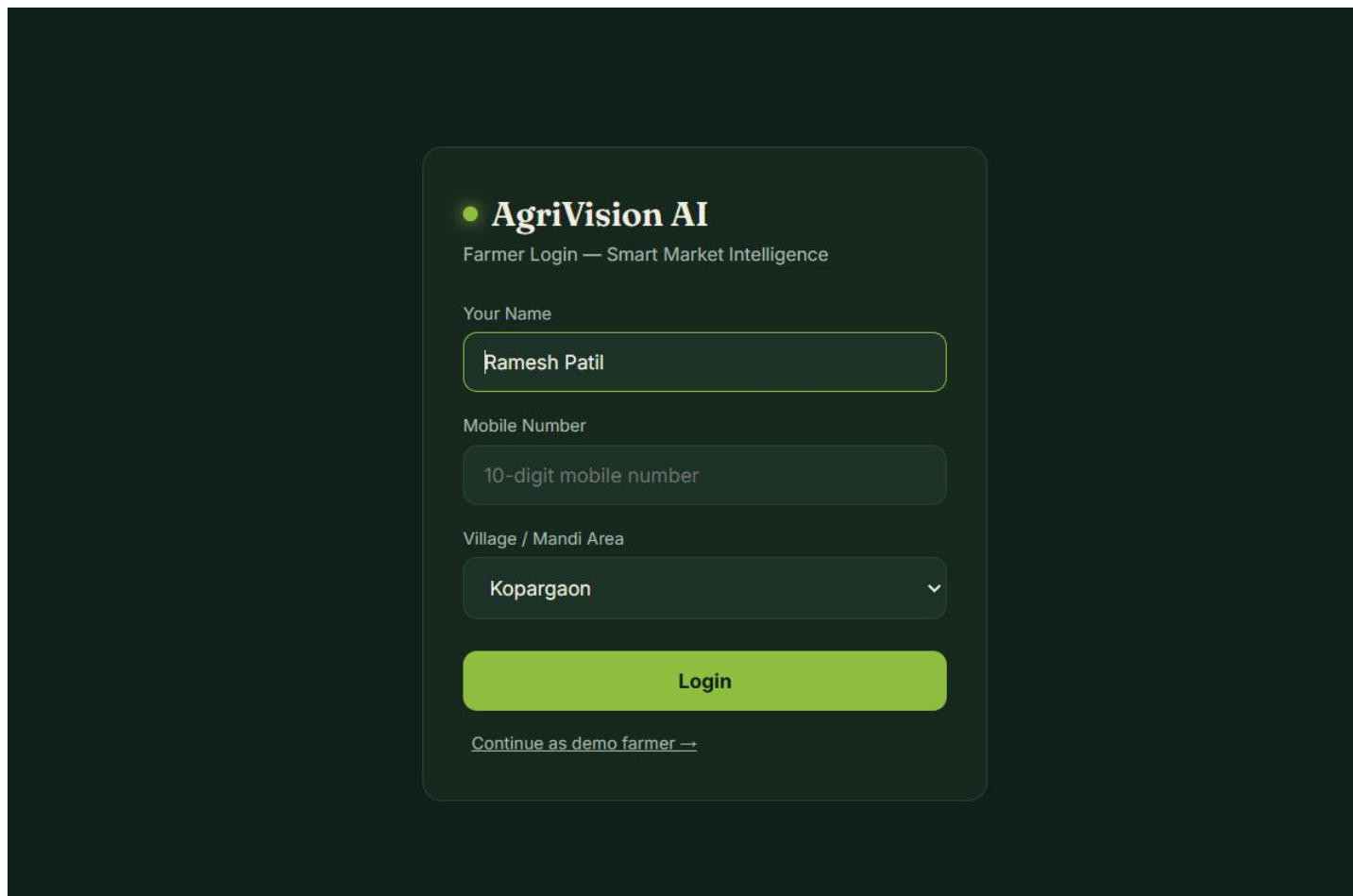
Future Technical Improvements

- Continuous retraining with model-drift monitoring.
- Benchmark ensembles empirically.
- Offline-first app and robust synchronisation.
- Personalised farmer recommendations.
- Production monitoring for forecast error, API health and recommendation outcomes.

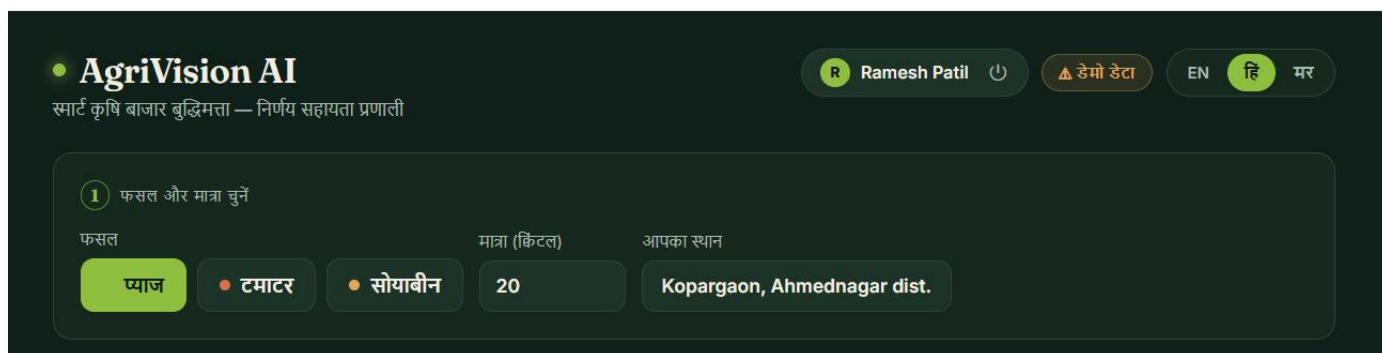
- **Device-Agnostic Access:** Extend the AI decision engine to deliver the same recommendations through **SMS, IVR calls, and missed-call services**, enabling farmers using basic phones to access AgriVision AI and reducing the **digital divide**.

Prototype Evidence

The supplied deck contains solution/USP and technical-approach visuals rather than a separate deployed-app screenshot; these should be described as design evidence, not deployment proof.



The image shows a dark-themed login screen for 'AgriVision AI'. The title 'AgriVision AI' is at the top, followed by the subtitle 'Farmer Login — Smart Market Intelligence'. Below this are three input fields: 'Your Name' with the text 'Ramesh Patil', 'Mobile Number' with the placeholder '10-digit mobile number', and 'Village / Mandi Area' with a dropdown menu showing 'Kopargaon'. A large green 'Login' button is positioned below the inputs. At the bottom, there is a link that says 'Continue as demo farmer →'.



The image shows the home screen of the 'AgriVision AI' application. At the top, the title 'AgriVision AI' is displayed, followed by the subtitle 'स्मार्ट कृषि बाजार बुद्धिमत्ता — निर्णय सहायता प्रणाली'. The user's name 'Ramesh Patil' is shown in a green circle, along with a 'डेमो डेटा' (Demo Data) button. The language is set to 'हि' (Hindi) and the region to 'मर' (Marathi). Below this, a section titled '1 फसल और मात्रा चुनें' (1 Crop and Quantity) is shown. It includes three buttons for crop selection: 'प्याज' (Onion), 'टमाटर' (Tomato), and 'सोयाबीन' (Soybean). The 'प्याज' button is selected. To the right, there are two input fields: 'मात्रा (किटल)' (Quantity in Kilo) with the value '20', and 'आपका स्थान' (Your Location) with the value 'Kopargaon, Ahmednagar dist.'.

- **AgriVision AI**

स्मार्ट कृषी बाजार माहिती — निर्णय सहाय्य प्रणाली

R Ramesh Patil



▲ डेमो डेटासेट

EN

हिं

मर

1 पीक आणि प्रमाण निवडा

पीक

कांदा

- टोमॅटो

- सोयाबीन

प्रमाण (किंटल)

20

तुमचे ठिकाण

Kopargaon, Ahmednagar dist.

② AI Recommendation

AI RECOMMENDATION

Medium confidence

WAIT ~5 DAYS

+ ₹878

Estimated extra profit by waiting, selling 20 quintals of Onion at Ahmednagar mandi instead of selling today.

BEST MANDI

Ahmednagar

48 km away

Price/quintal

Transport

Net total

₹2,183 -₹86 ₹41,925



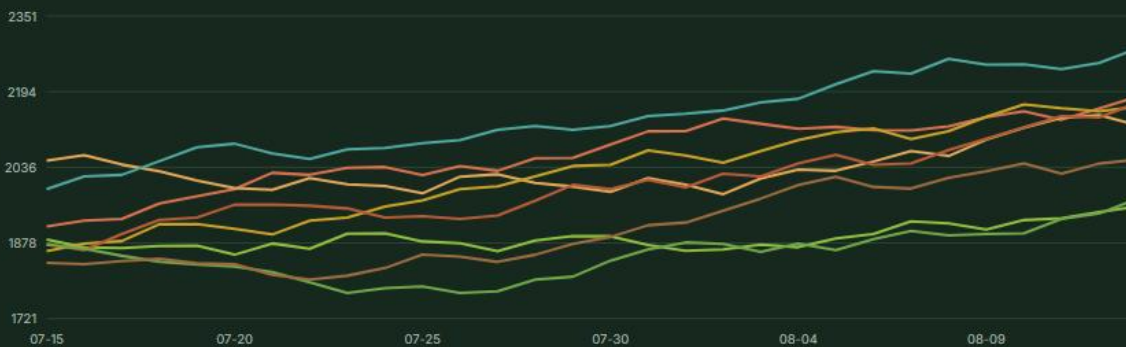
WEATHER & SEASONAL INSIGHT

Why: Festival demand rising — Ganesh Chaturthi & Diwali season ahead. Mandis historically see an 8-12% price increase in this window.

Dry spell across the Nashik belt (Lasalgaon) has slowed fresh arrivals, tightening supply and supporting the price rise.

3 30-Day Price Trend

■ Kopergaon
 ■ Ahmednagar
 ■ Nashik
 ■ Chh. Sambhajinagar
 ■ Pune
 ■ Solapur
 ■ Kolhapur
 ■ Nagpur



4 Mandi Comparison — Ranked by Your Net Profit

BEST PICK

1	Ahmednagar 48 km away	Price/quintal ₹2,183	Transport -₹86	Net total ₹41,925
2	Pune 180 km away	Price/quintal ₹2,283	Transport -₹234	Net total ₹40,977
3	Nashik 85 km away	Price/quintal ₹2,124	Transport -₹136	Net total ₹39,761
4	Chh. Sambhajinagar 120 km away	Price/quintal ₹2,162	Transport -₹180	Net total ₹39,639
5	Kopargaon 5 km away	Price/quintal ₹1,954	Transport -₹11	Net total ₹38,860
6	Solapur 220 km away	Price/quintal ₹2,170	Transport -₹264	Net total ₹38,111
7	Kolhapur 370 km away	Price/quintal ₹1,970	Transport -₹407	Net total ₹31,258
8	Nagpur 590 km away	Price/quintal ₹2,052	Transport -₹531	Net total ₹30,417

Demo Dataset — Onion

This is the exact synthetic dataset powering the demo — 30 days of price (₹/quintal) across 8 Maharashtra mandis, modeled on real Agmarknet patterns. Swap this file for a live Agmarknet/eNAM feed in production.

Date	Kopargaon	Ahmednagar	Nashik	Chh. Sambhajinagar	Pune	Solapur	Kolhapur	Nagpur
2026-07-15	1885.95	1912.56	2049.93	1861.60	1990.39	1876.23	1874.49	1837.20
2026-07-16	1869.16	1924.69	2061.41	1877.03	2017.17	1863.36	1866.24	1834.29
2026-07-17	1867.94	1928.66	2041.51	1882.32	2020.17	1896.91	1851.74	1840.32
2026-07-18	1872.23	1960.75	2027.54	1917.26	2048.80	1926.46	1839.18	1845.15
2026-07-19	1872.36	1975.65	2008.17	1918.00	2077.40	1931.27	1833.55	1835.80
2026-07-20	1853.79	1990.63	1992.59	1907.64	2085.16	1957.90	1829.25	1834.30
2026-07-21	1877.40	2024.45	1989.17	1895.92	2084.80	1958.51	1817.53	1811.84
2026-07-22	1866.37	2020.15	2013.48	1924.53	2053.08	1955.74	1796.30	1802.21
2026-07-23	1897.61	2034.88	2000.70	1931.04	2073.15	1950.30	1774.44	1810.22
2026-07-24	1898.30	2036.48	1997.12	1954.25	2076.12	1930.93	1784.37	1826.09
2026-07-25	1881.65	2019.83	1981.72	1967.09	2086.31	1933.92	1787.49	1853.98
2026-07-26	1877.79	2038.46	2016.52	1990.59	2092.48	1928.34	1773.92	1849.88
2026-07-27	1861.07	2028.80	2021.47	1996.58	2114.39	1935.61	1777.72	1838.93

Download full dataset (CSV)

8. References

Government / Data

- Agmarknet: <https://agmarknet.gov.in/home>
- data.gov.in: <https://data.gov.in/>
- eNAM: referenced as a market-data source in the project technical approach.

Institutional

- NABARD: <https://www.nabard.org/>

Frameworks / APIs

- Flutter; FastAPI; PostgreSQL; XGBoost; Prophet; scikit-learn; OpenWeather API; Google Maps API; Firebase Cloud Messaging; JWT; Google Translate API.

Project Source

- Smart Kopargaon Hackathon, Problem Statement skh005, Smart Agricultural Market Intelligence System; Team The Elite Four, Team ID 8K5YVT.

Research Findings from Submitted Deck

- Market information affects selling decisions; crop prices are influenced by demand, weather and seasonal trends; mandi prices can differ for the same crop; AI/ML forecasting is proposed; multilingual/voice interfaces improve rural accessibility.

Documentation constraint: exactly 8 pages; concise technical content only. Unsupported accuracy values and production claims are intentionally excluded.